

ASTROID: Adhesive Stamping Technology for Regolith

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Abstract

The Columbia Space Initiative has designed the Columbia ASTROID, a contact sampling device intended to pick up the top 1-5mm of lunar regolith while maintaining the grain orientation of the top exposed layer of surface particles for geological surveying and analysis back on Earth. The device features two detachable components: the Stamping Device and the handle. The Stamping Device is modeled after a self-inking stamp and is what is utilized to actually take the regolith sample, using an adhesive-based Sample Plate. The handle is used to apply an axial force to actuate the device from a comfortable height and position for the user. Testing in NASA's Simulant Development Laboratory (SDL) has confirmed ASTROID's ability to maintain grain orientation, however, some areas for improvement to address in future iterations were identified during testing in the Neutral Buoyancy Lab (NBL) pool-side testing environment, in regards to operator usability and friction issues related to lack of dust mitigation.

Introduction

NASA's Artemis III mission aims to collect undisturbed lunar regolith samples from the Moon's south pole—an area less studied than other regions. A key scientific objective is to analyze the top 1–5 mm of regolith to better understand space weathering and the Moon's geological history. However, current methods fail to preserve the grain orientation of these delicate surface layers. To address this, the ASTROID (Adhesive Stamping Technology for Regolith Orientation Integrity Delineation) device was developed. It uses a stamping mechanism and a lab-engineered adhesive gel to collect surface regolith while maintaining its orientation. The adhesive cures into a flexible, adherent substance that dissolves with heptane, ensuring sample stability during extraction and transport.

Designed for use in the harsh conditions of the lunar south pole—where temperatures are extremely low and the regolith is fine-textured—the ASTROID system emphasizes reliability, ergonomic operation in EVA gloves, and secure sample containment. Its design draws inspiration from tools used in missions like Apollo and OSIRIS-REx, while introducing improvements in usability and sample integrity. The device has undergone functionality testing in NASA's Neutral Buoyancy Laboratory and sample performance testing in the Simulant Development Lab. Its self-inking stamp-inspired mechanism and easily removable sampling plate ensure both ease of use and secure storage, supporting Artemis III's goal of returning scientifically valuable samples to Earth.

Design Overview

The ASTROID is a fully manual device that uses a self-inking stamp-style mechanism to collect contact samples of lunar regolith. Instead of ink, it uses a custom adhesive Sample Plate to embed and preserve surface grains. The system is composed of two detachable components: the Handle and the Stamping Device. The Stamping Device includes four primary parts—the Stamp Track, Stamp Cap, Carriage, and Sample Plate—which work together to guide, flip, and secure the Sample plate during actuation.

The Stamp Track is a five-sided box open at the bottom to interface with the lunar surface, featuring cutouts on two opposing vertical walls to guide the Carriage prongs. It also houses the winch system that holds the constant force spring system which suspends the Stamp Cap, a complementary five-sided cover, in the unactuated position over the Stamp Track until a downward force is applied via the Handle. The Carriage holds the Sample Plate and performs

the flipping motion typical of a stamp; it includes two prongs that extend from the sides of the Carriage through the Stamp Track cutouts and into slotted holes in the Stamp Cap. These slotted geometries constrain the Carriage to follow a combined translational and rotational path. The Sample Plate itself consists of a PLA base bonded to an acrylic plate that the adhesive is applied to, forming the collection surface that contacts and preserves the regolith.

Proposed Design

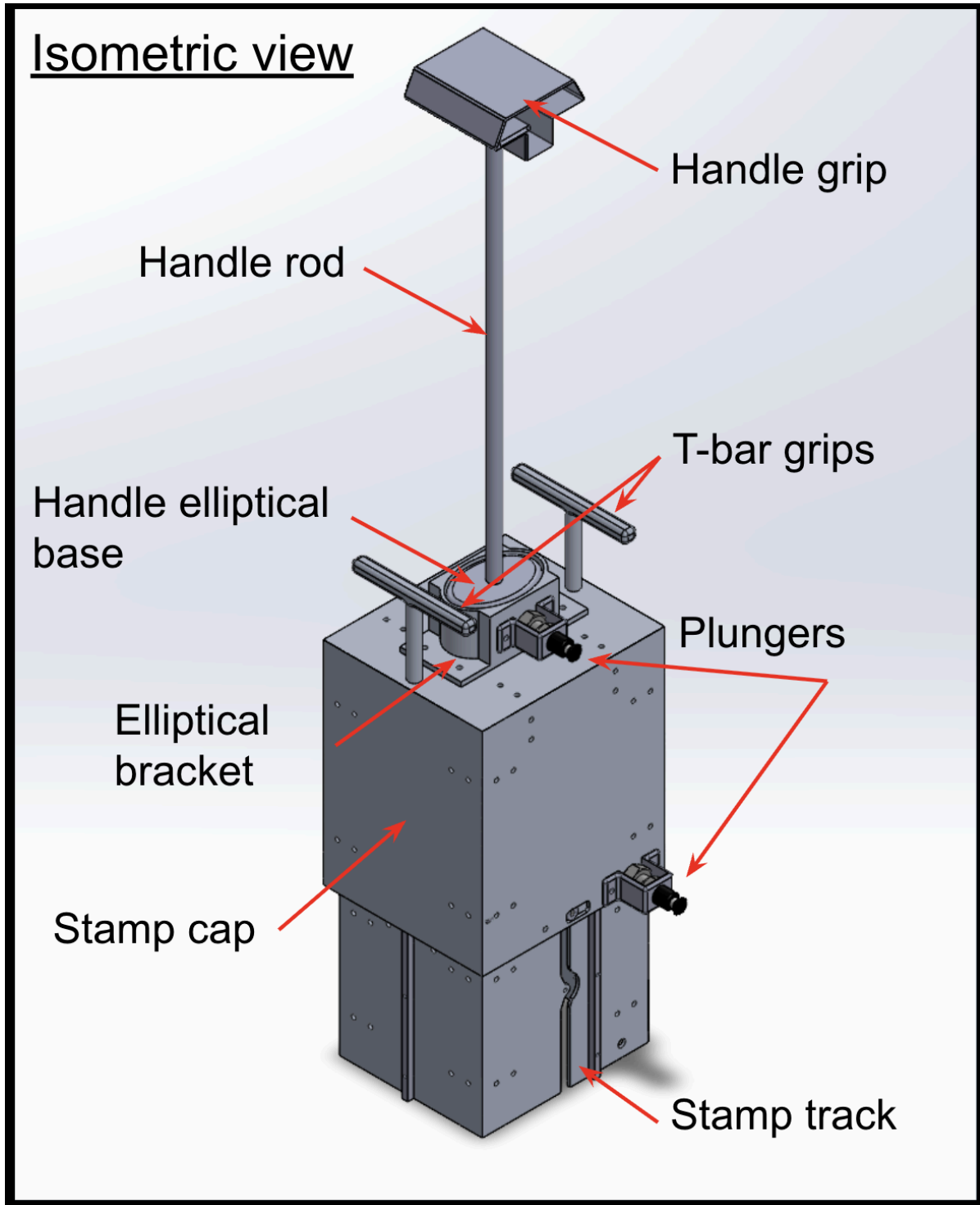
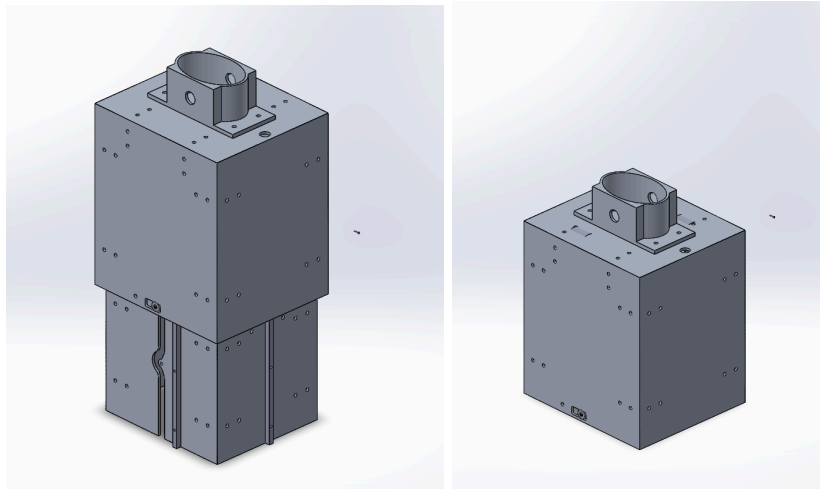
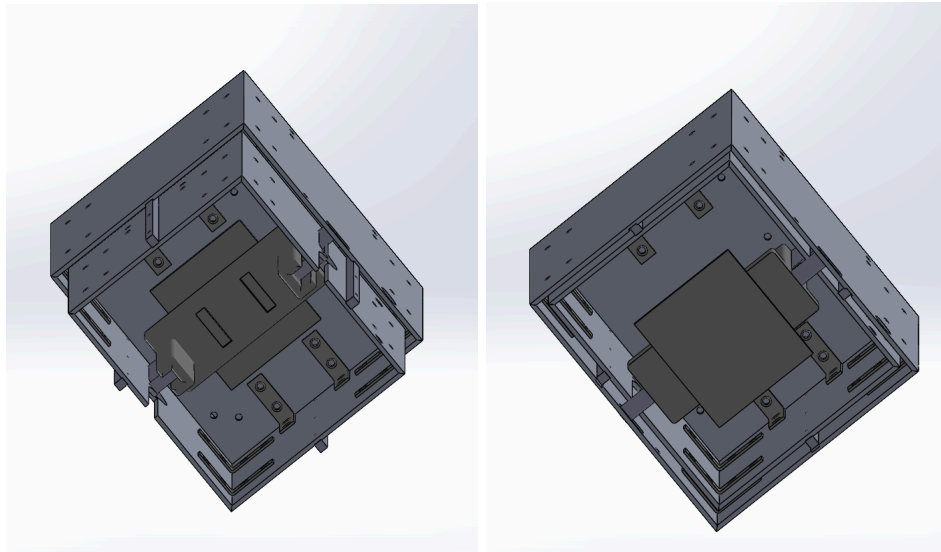


Figure 1. Labeled image of the ASTROID Device: External View



**Figure 2. Stamping Device's Four Subcomponents: Exterior View
Unactuated (left) Actuated (right)**



**Figure 3. Stamping Device's Four Subcomponents: Interior View
Unactuated (left) Actuated (right)**

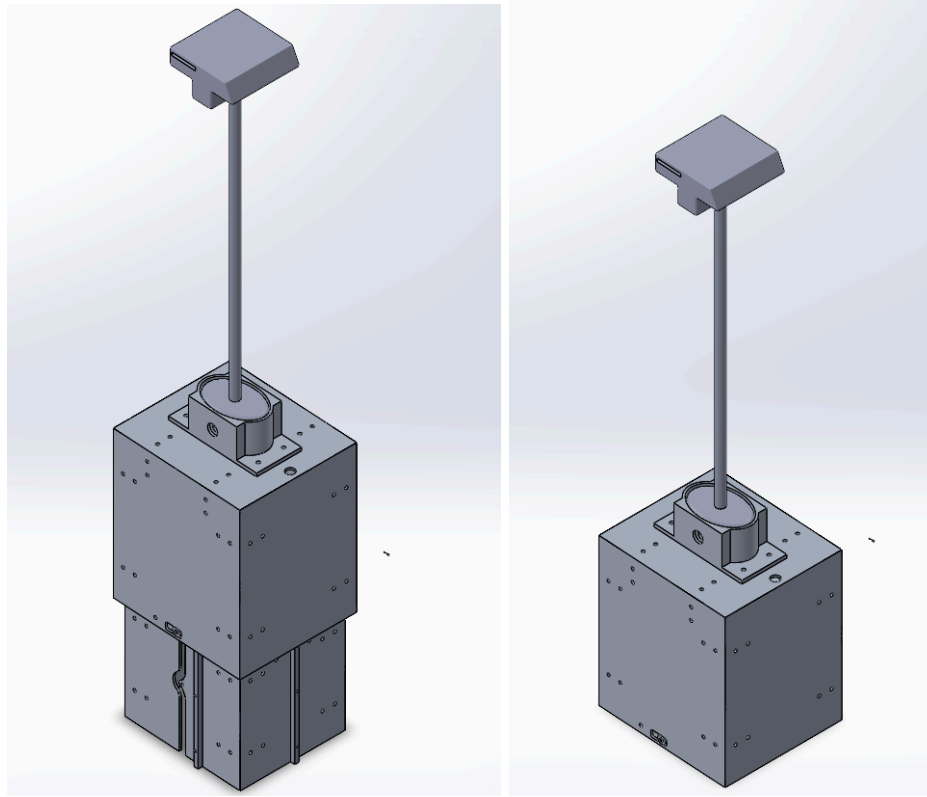


Figure 4. ASTROID in Unactuated (left) and Actuated Positions (right)

Dimensions

Total Weight: Primary configuration: 12 lbs

Component	Weight (oz)	Dimensions (in)	Materials
Stamp Cap	3.81lbs	9.2" x 7.9" x 6.9"	6061 Aluminum Alloy
Carriage	0.44lbs	5" x 3.8" x 2"	Performance PLA
Carriage Assembly	0.48lbs	6.89" x 3.8" x 2"	Performance PLA, Brass/Steel Hardware
Locking Mechanism	2.4oz	2.7" length 0.6" diameter	1045 Carbon Steel, Polyacetal
Track (Top)	8 oz	7.3"x6.3"x0.125"	Aluminum Alloy
Track (Right)	10.08 oz	7.3"x6.3"x0.125"	Aluminum Alloy
Track (Left)	10.08 oz	7.3"x6.3"x0.125"	Aluminum Alloy
Track (Front Back)	9.12 oz	6.3"x7.3"x0.125"	Aluminum Alloy
Track Prong	0.53 oz	Length: 0.9", Head: diameter 0.36"	Aluminum Alloy
Split Track Assembly with	12 x brackets + Assembly	7.3"x7.3"x6.3"	Aluminum Alloy

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Brackets			
Cap Top	0.64lbs	7.9"x6.9"x0.125"	6061 Aluminum Alloy
Cap Left/Right	13.44oz	9.2"x7.9"x0.125"	Aluminum Alloy
Cap Front/Back	11.84 oz	9.2"x6.9"x0.125"	Aluminum Alloy
Split Cap Assembly w/ Brackets	12 x brackets + Assembly	9.82"x7.9"x6.93"	6061 Aluminum Alloy
Heat-Set Insert	.06 oz	Thread Length: 19.05mm	Brass
Set Screw	0.24 oz	Length: 2" Thread: ¼"	Alloy Steel
Steel Socket Head Screw	0.003 oz	Length: ¼ ", Head: diameter 0.096", Height 0.06"	Black Oxide Alloy Steel
Bracket	0.42 oz	1.5"x1.5x0.5"	Galvanized Steel

Stamping Device

Purpose:

The stamping component of the ASTROID device collects a 1–5 mm sample of lunar regolith using a single linear, glove-compatible motion. This mechanism distributes contact force evenly to gather and store a sample without disturbing its orientation. It eliminates the need for multiple tools or steps, reducing astronaut fatigue and complexity during EVA.

Design Overview:

The Stamping Device includes the Stamp Cap, Stamp Track, Carriage, and Sample Plate. When axial force is applied, the Carriage follows a path guided by two sets of tracks—one curved and one straight—that flip the Sample Plate 180° during its downward motion. A spring resets the Carriage to its stowed position, preserving the sample's surface orientation by pressing it into the ceiling of the Stamp Track. The sample area measures 3" x 3" within a 3.2" x 3.6" plate and contains the adhesive for regolith collection.

Handle

Purpose:

The handle enables astronauts to actuate the Stamping Device with minimal hand motion, a critical consideration for EVA glove use. Rounded rods were avoided in favor of ergonomic features that reduce fatigue and maximize surface contact for palm-based pressure.

Design Overview:

The grip includes a triangular lip and recess optimized for EVA gloves, following NASA's Man-Systems Integration Standards. It connects to the Stamp Cap via a concentric elliptical base and a locking mechanism. An internal aluminum rod connects the grip to the base, resulting in a total height of 20.8".

Plunger Mechanism

Purpose:

The plunger mechanism secures device components by sliding a pin from one face into a hole on a parallel face with a button press, locking them in place and preventing unintended motion during use.

Design Overview:

The ASTROID device uses three locking points to secure key components during operation and stowage. Two plungers are located on the elliptical bracket to attach or detach the handle from the Stamping Device. These allow

the handle to be locked in place for use and unlocked for stowage. The third plunger is located on the Stamp Cap and is used to lock the Stamping Device in either the actuated or unactuated position—ensuring the Sample Plate remains secure during storage and enabling safe attachment or removal of the Sample Plate from the Carriage when ready to analyze.

Sample Slide Detachment

Purpose:

This mechanism allows the Sample Plate to be easily removed from the Carriage once testing is complete, enabling safe regolith analysis at the Simulant Development Lab (SDL).

Design Overview:

After sampling, the device remains locked in its actuated position to keep the Sample Plate exposed for retrieval. The set screws—also serving as the Carriage prongs that dually connect the Sample Plate to the Carriage—are loosened using a hex key. The design provides adequate finger clearance beneath and around the plate, with built-in safety margins to minimize contamination risk and prevent finger entrapment. This allows for the careful and secure removal of the adhesive-embedded sample.

Adhesive

Purpose:

The adhesive captures and secures fine regolith particles, preserving their surface orientation for analysis at the SDL.

Design Overview:

3M™ Tape 665 adhesive, stripped with heptane and dried in layers, forms a 500-micron pad that embeds particles via applied force. It is non-water-soluble and temperature resistant from -60°F to 125°F, meeting NBL and SDL standards. This technique has been used effectively in past biological experiments, such as embedding fruit fly embryos, demonstrating its suitability for lunar simulants of a relatively similar dimensional magnitude.

Device Testing

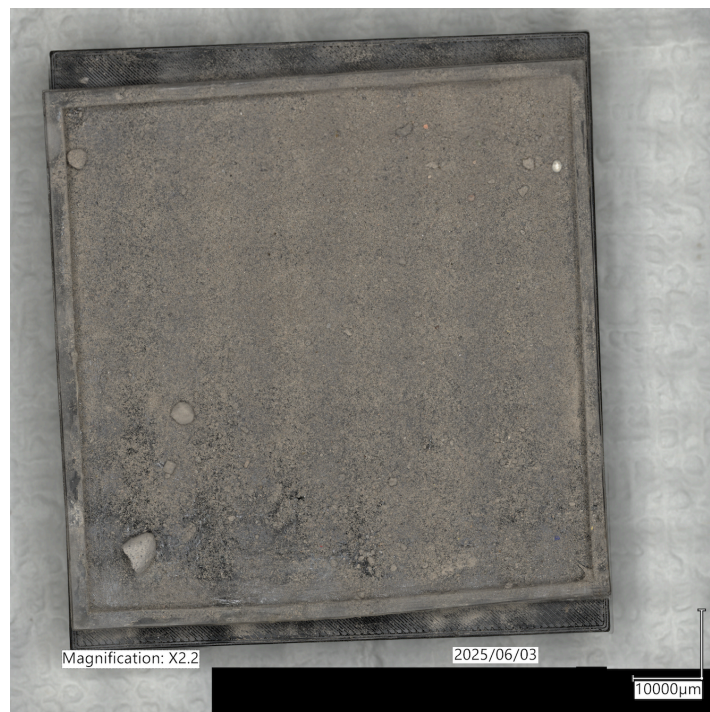


Figure 5. NASA Simulant Development Laboratory. Official testing conducted June 2, 2025.

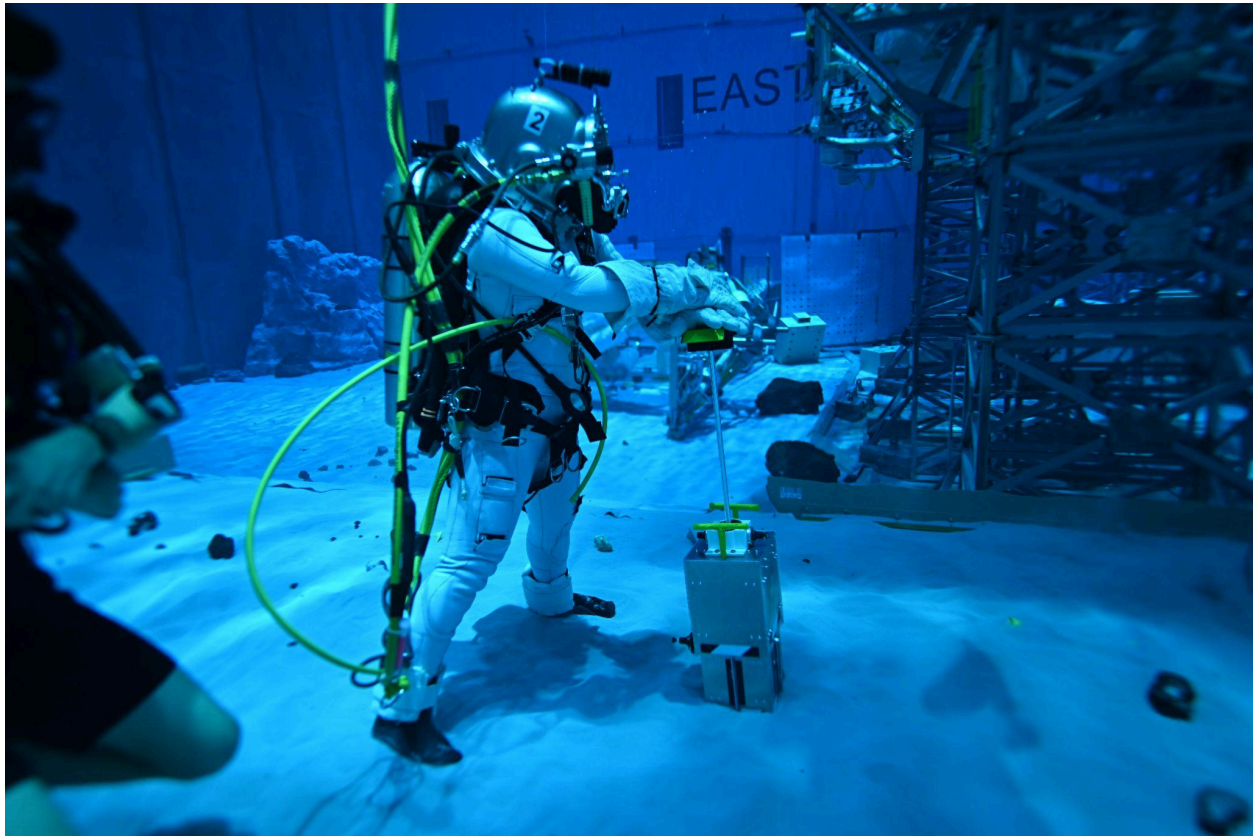


Figure 6. NASA Neutral Buoyancy Laboratory. Official testing conducted June 3, 2025.

Testing Objectives

Substantial consideration was given to establishing meaningful testing objectives for the ASTROID device prior to its evaluation at the NBL and SDL. Our team operated the ASTROID device at the Simulant Development Lab (SDL), while a diver unfamiliar with the device's construction operated it during testing at the Neutral Buoyancy Lab (NBL). The two primary goals were: (1) to validate the device's ability to collect a simulant-based sample while preserving surface grain orientation at the SDL, and (2) to assess the device's usability in simulated microgravity conditions at the NBL. The team prioritized evaluating the smoothness of the stamping mechanism and the adhesive's effectiveness in regards to embedding regolith. To assess grain orientation preservation, a photo of the sampling region was taken before actuation and later compared to a microscope image of the Sample Plate post-collection.

To quantify usability, participants were asked to rate various aspects of the device and their experience on a scale from 1 to 5, where 1 indicated the feature failed completely and 5 indicated flawless performance. Maintaining a consistent scale was essential for comparing results across different categories. A sample testing procedure is included below, highlighting how the numerical scale was integrated into a clear, step-by-step protocol to support a smooth and intuitive user experience.

SDL Operation Plan

*Beginning with the Handle and Stamping Device separated from one another. This operation plan is executed directly by the Columbia Space Initiative team.

A. Device Assembly

1. Holding the Handle by the aluminum rod and the Stamping Device by the t-bar handles, place the Elliptical Base of the Handle inside of the Elliptical Bracket on the Stamp Cap (orientation does not matter).
2. Lock the plungers on the side of the Elliptical Bracket to securely attach the Handle to the Stamping Device. While holding the Stamping Device by the t-bars, ensure that the white portions of the plunger are no longer visible and lightly tug on the handle with one hand to ensure locking was successful.
3. Return to grasping the ASTROID by both of the yellow t-bars on the Stamping Device.

B. User Instructions

1. Subject is given the ASTROID with both the Stamping Device and detachable handle already assembled in the locked, unactuated position. Hold by both of the indicated yellow T-bars.
2. Subject ensures both the plungers on the handle and the plunger on the bottom of the cap are locked.
3. Subject uses both T-bars to lift the Stamping Device and places the assembled device onto a relatively flat area of regolith. The opening on the bottom of the Stamp Cap should face the surface.
 - a. Note that the direction of the vertical device does not matter so the subject can reposition the device by slowly lifting directly up and rotationally reorienting, while trying to minimize surface disturbance.
4. Subject squats down to unlock the plunger on the side of the Stamp Cap to allow actuation for sample collection. Once complete, the subject returns to a comfortable standing position.
5. Once standing, the subject wraps fingers around the triangular lips of the handle. From there, the palms should sit flat on the top surface of the handle. The subject's wrists should meet in the middle.
6. Subject slowly actuates the device by pressing downward on the handle. Continue to apply pressure until the cap of the Stamping Device has made contact with the surface of the simulant, or when the device is no longer able to be pushed any further.
 - a. If the cap stops moving down before the bottom of the cap makes contact with the regolith (i.e. the cap doesn't completely cover the inner Stamp Track), the subject should hold the top of the handle with one hand and try to bring the device back up to the unactuated position by gently shaking and applying an upward force on the sides of the cap with the other hand. Then the subject can try stamping the device down again.
7. The subject holds pressure in place for 3 seconds when at the bottom of the actuation, then releases pressure slowly and in a controlled fashion to prevent a fast/uncontrolled recoil, allowing the device to restore itself to the unactuated position.
 - a. If the cap can not move back up, the subject should put one hand on the top of the handle, and with the other handle gently shake the side of the device. Try to minimize dust disturbance on the surface. Apply more pressure until it works.
8. Once the device returns to the unactuated position, the subject uses the plunger on the Stamp Cap to lock it in the unactuated position.

C. Sample Plate Detachment and Analysis

9. Subject now performs steps for Device Assembly in reverse to free the Stamping Device from the Handle.
10. Once separated, the subject places the Stamping Device on its side with the Stamp Cap's plunger facing upwards towards the ceiling. The subject can unlock the plunger.
11. This next step requires two subjects. One subject will position themselves facing the Elliptical Bracket on the top of the Stamp Cap and the other subject will face the open end of the Stamping Device.
12. Both subjects will push to actuate the device in order to expose the used sample plate. The subject pushing from the open end of the Stamping Device needs to use flat, open hands to avoid pinch points.
13. Once the sample plate is exposed, one subject can lock the plunger in the actuated position.
14. From there, the device is fixed in this position and the entire Stamping Device can now be flipped upside down, resting on the Elliptical Bracket.

15. Now, one subject should securely hold the Stamping Device by its sides, while the other uses a 2mm allen key to unscrew the set screw prongs of the Carriage with one hand and holds the bottom of the Carriage body with the other hand. Repeat this process for both set screws.
16. Once free from the rest of the Stamping Device, the Carriage and Sample Plate should be separable by using the overhanging lip of the Sample Plate to free it from the Carriage rectangular cut-outs.
17. The Sample Plate is ready for analysis.

NBL Operation Plan

*Beginning at the test surface level of the Neutral Buoyancy Laboratory while the Columbia Space Initiative team is in the Control Center directing the diver with the steps below.

A. General Disclaimers

1. Before use, verify that the device is in the locked, unactuated position.
2. Plunger Mechanism Function:
 - a. Unlocked: White part is visible on plunger (external portion of plunger is extended).
 - b. Locked: White part not visible on plunger (external portion of plunger is retracted).
 - c. Press the plunger to toggle between states.
 - d. Plungers on the handle do not need to be pressed simultaneously.
3. Set screw prongs on the Carriage should be properly aligned through the slots on the Stamp Cap.
4. Surface placement should be on flat ground with pebbles no larger than 2 mm in diameter (reference marked in red on the Stamp Cap).

B. Diver Instructions

18. Subject is given the ASTROID with both the Stamping Device and detachable handle already assembled in the locked, unactuated position. Hold by indicated yellow T-bars.
19. Ensure both the plungers on the handle and the plunger on the bottom of the cap are locked.
20. Subject uses both T-bars to lift the Stamping Device and places the assembled device onto a relatively flat area 2 mm, a reference of this diameter is marked in red on the top of the Stamp Cap. The opening on the bottom of the Stamp Cap should face the surface.
 - a. Note that the direction of the device does not matter so the subject can reposition the device by slowly lifting directly up and reorienting, just try to minimize surface disturbance.
21. Subject squats down to unlock the plunger on the side of the Stamp Cap to allow actuation for sample collection. Once complete, return to a comfortable standing position.
22. Once standing, the subject wraps fingers around the triangular lips of the handle. From there, the palms should wrap around and be flat on the top surface of the handle. The wrists should meet in the middle.
23. Subject slowly actuates the device by pressing downward on the handle. Continue to apply pressure until the cap of the Stamping Device has made contact with the surface of the simulant, or when the device is no longer able to be pushed any further.
 - a. If the cap stops moving down before the bottom of the cap makes contact with the regolith (i.e. the cap doesn't completely cover the inner Stamp Track), hold the top of the handle with one hand and try to bring the device back up to the unactuated position by gently shaking and applying an

upward force on the sides of the cap with the other hand. Then try stamping the device down again.

24. Hold pressure in place for 3 seconds when at the bottom of the actuation, then release pressure slowly and in a controlled fashion to prevent a fast/uncontrolled recoil, allowing the device to restore itself to the unactuated position.
 - a. If the cap can not move back up, put one hand on the top of the handle, and with the other handle gently shake the side of the device. Try to minimize dust disturbance on the surface. Apply more pressure until it works.
25. Once the device returns to the unactuated position, use the plunger on the Stamp Cap to lock it in the unactuated position.
26. Using the T-bars, lift the ASTROID and return to the water's surface.

C. Performance Questions

1. On a scale of 1 to 5, with 1 being the worst and 5 being the best, how would you rate the overall performance of the device?
2. On a scale of 1 to 5, with 1 being very little and 5 being too much, how would you rate the amount of force required to actuate the device?
3. On a scale of 1 to 5, with 1 being easy and 5 being impossible, how would you rate the difficulty of locking and unlocking the plunger mechanisms? Does the device return easily to the unactuated position?
4. On a scale of 1 to 5, with 1 indicating it gets easier and 5 indicating it gets harder, how would you rate the difficulty of actuating the device multiple times?
5. On a scale of 1 to 5, with 1 being the most comfortable and 5 being the least comfortable, how would you rate the ergonomics of the handle and its grip? Similarly, how would you rate the t-bars?

Results

During the SDL test, successful results were obtained after a single actuation of the ASTROID device into the simulant sampling region. The resulting Sample Plate was analyzed under a microscope and confirmed that the adhesive gel preserved surface grain orientation during collection, validating its initial performance. However, after three successive actuations using the same device, frictional issues developed between the Stamp Cap and Stamp Track. These were caused by regolith particles accumulating along the inner interface of the Stamp Cap, resulting in increased resistance and eventually locking the mechanism. This indicates that the device reaches a physical sampling limit after multiple uses, likely due to insufficient clearance between the two parallel plates that slide against each other during actuation.

During the NBL test, successful sampling was achieved following explanation of the usage procedure and rating scale. While there were no issues during setup, the user encountered operational challenges due to similar frictional resistance caused by sand entrapment at the bottom of the pool. Feedback from the diver emphasized the need to mitigate dust and sand buildup in the narrow space between the Stamp Track and cap. Although actuation on a solid surface was relatively easy for non-expert users, the underwater friction led to a lower usability score. Additional concerns were raised about potential pinch points, especially during troubleshooting. All 12 minutes of NBL testing were needed to overcome resistance and successfully actuate the device, which ultimately required tilting it sideways and applying force to both ends. This method introduced a risk of finger entrapment when users grasped the bottom of the Stamp Track during troubleshooting. Additionally, the team experimented with using lubricants to reduce friction between components; however, this approach proved counterproductive. The lubricant caused the sand and simulant to adhere more readily to internal surfaces, exacerbating resistance and making actuation more difficult.

Future Iterations

Improved Internal Clearances and Dust Mitigation:

One of the most pressing issues observed during both the SDL and NBL tests was the buildup of regolith or sand between the Stamp Cap and Stamp Track, resulting in friction and mechanical lock-up after repeated actuations. Future iterations of the ASTROID device should prioritize increasing the clearance between these sliding components. While a tight interface helps maintain alignment, the current tolerance appears too narrow to account for particulate contamination. A redesign incorporating wider gaps or integrated dust channels could help displace or trap regolith before it accumulates in critical sliding areas. Additionally, the application of low-friction, dust-resistant coatings or materials (e.g., PTFE-lined tracks) could significantly reduce the impact of embedded particles.

Reevaluation of Lubrication Strategy:

The test results showed that adding lubricants led to increased sand and simulant adhesion, ultimately worsening resistance. A future version of the ASTROID device should avoid external lubricants and instead consider self-lubricating materials such as UHMW polyethylene or dry-film coatings like molybdenum disulfide. These materials maintain low-friction properties without attracting particulates, thus preserving mechanical smoothness while operating in abrasive environments.

Conclusion

Our team designed the ASTROID system to enable astronauts to safely and effectively collect scientifically valuable lunar regolith samples during Artemis III, preserving the delicate surface grain orientation critical to understanding space weathering and the Moon's geological history. By combining a self-inking stamp-inspired mechanism with a custom adhesive and ergonomic EVA-compatible features, ASTROID offers a simple, single-motion sampling process suited to the harsh conditions of the lunar south pole.

During our testing at NASA's Simulant Development Lab and the Neutral Buoyancy Laboratory, we gained valuable insights into the device's strengths and areas for refinement. At the SDL, ASTROID successfully preserved surface orientation and collected high-fidelity samples in a single actuation. At the NBL, we confirmed that the device could be operated by users unfamiliar with its internal mechanisms, though friction and dust-related interference presented usability challenges. Feedback from the NBL diver helped identify critical improvements to internal clearances, dust mitigation strategies, and ergonomic safety.

These results guide future iterations aimed at increasing durability and reliability across repeated actuations. With continued development, ASTROID will be well-positioned to support NASA's goal of returning pristine, undisturbed samples from the lunar surface—advancing our understanding of the Moon and laying the groundwork for future planetary exploration.

Acknowledgments

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